Chapter 1

ICT Fundamental

Student Learning Outcomes (SLOs):

By the end of this unit, students will be able to:

- Explain and analyze how emerging technologies, including Artificial Intelligence (AI), 5G, robotics, computer-assisted translation, 3D and holographic imaging, virtual reality, distributed applications, blockchain, and machine learning, are applied across various fields and industries. They will gain insights into the unique roles and impact of each technology in everyday life and professional contexts.
- Identify the structure and purpose of a network and analyze its core components, including routers, switches, and hubs.
- Describe Software Networking component with reference to their use.
- Recognize and understand the network security issues including firewall and its application.

1.1 Introduction to Unit:

Information and Communication Technology (ICT) is a key part of our daily lives, helping us at home, school, and work. This unit will teach students how ICT makes tasks simpler, such as using computers and the internet. Students will also discover exciting topics like Artificial Intelligence (AI), the Internet of Things (IoT), and emerging technologies. By the end of the unit, they will learn to use technology smartly and think of innovative ideas for the future.

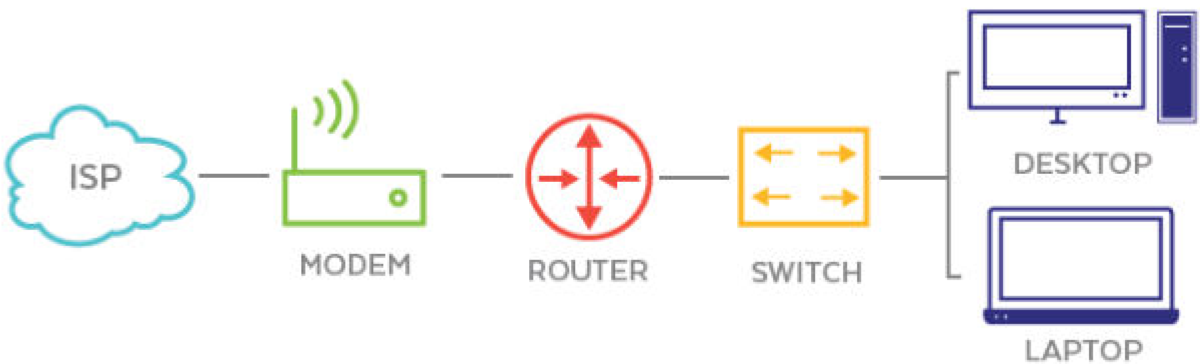


Fig 1.1 Network Diagram

1.2 Impact of Emerging Technologies:

Emerging Technologies are new advancements in information and communication technology (ICT) that improve how we live, work, and learn. They make everyday tasks easier and more efficient.

1.2.1 Internet of Things (IoT):

The Internet of Things (IoT) is a system where everyday devices are connected to the internet. These devices can talk to each other and be controlled remotely.

Examples:

- **Refrigerators:** A smart fridge can tell you what food items you need to buy and even order them online.
- **Cars:** Modern cars can use the internet to provide navigation help, detect maintenance issues, and even drive themselves in some cases.

- **Air Conditioners:** You can adjust the temperature of your home using your phone, even when you're not there.

IoT makes devices smarter and more convenient, allowing for better control and automation in our daily lives.

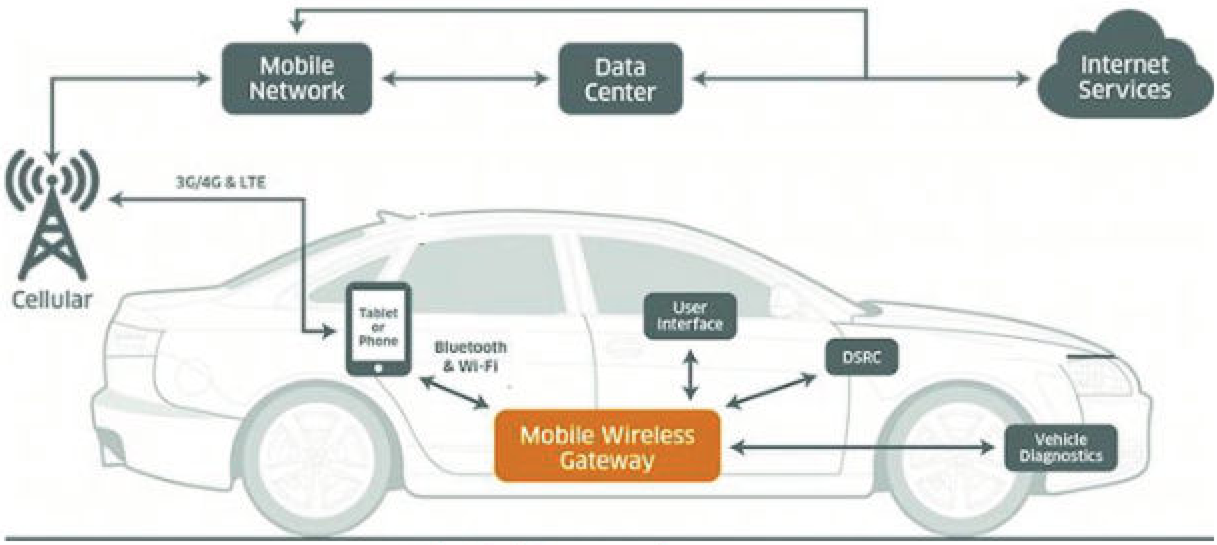


Fig 1.2.1 Connected car system

1.2.2 Embedded Systems:

An embedded system is a small computer inside a larger device, designed to perform one specific task. These systems are found in many everyday items and help them run smoothly.

Examples:

- **Cars:** Embedded systems control things like the engine, airbags, and entertainment systems.
- **Home Appliances:** Washing machines and refrigerators use embedded systems to set timers and keep temperatures just right.
- **Medical Equipment:** Devices like MRI machines and heart monitors rely on embedded systems to work accurately.
- **Phones and Internet:** Networking devices use embedded systems to manage data and communication.

- **Military Equipment:** Drones and navigation systems are controlled by embedded systems.

Embedded systems make these devices more efficient and reliable.

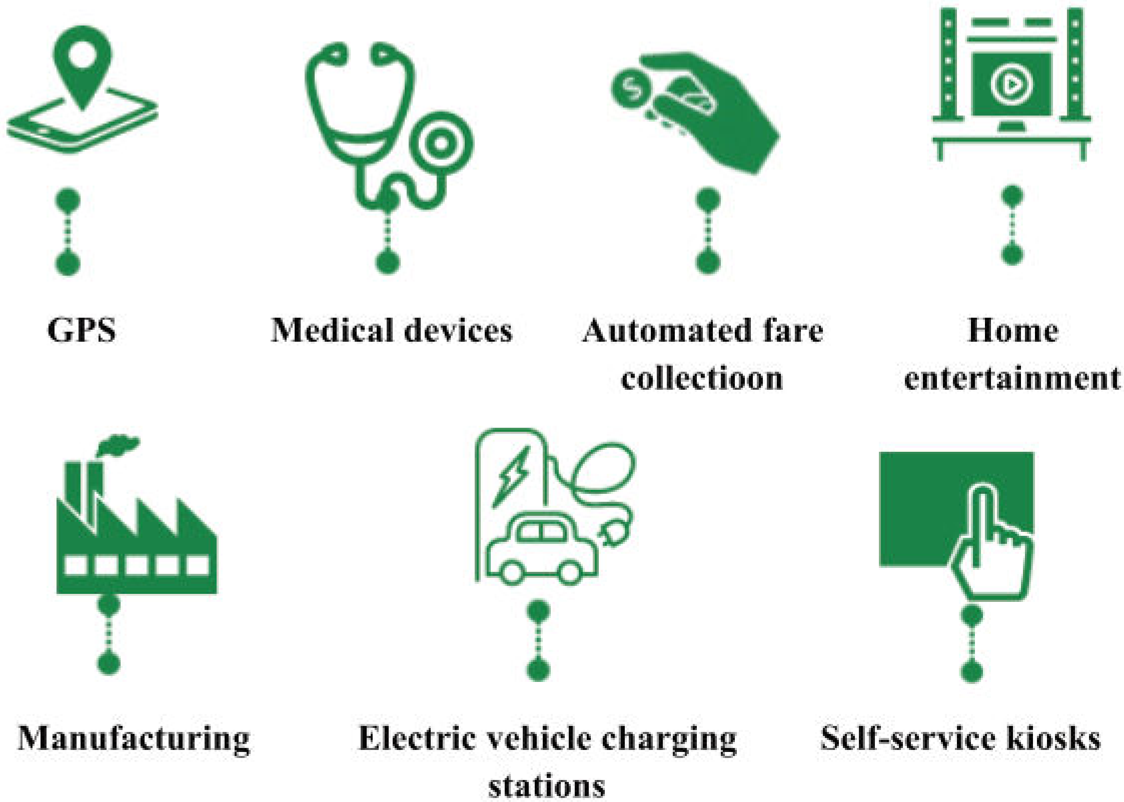


Fig 1.2.3 Applications of Embedded Systems

1.2.3 Edge Computing:

Edge computing is like having a mini-computer close to the device or sensor where data is generated. Normally, data is sent to a centralized system, a remote server that handles it. While it is effective but system can be slow due to long data travel distances. Edge computing solves this by processing data locally, making the system faster and more efficient.

Edge computing in cars helps to detect the traffic in real-time and quickly suggests a faster route quickly to avoid heavy traffic. It can also detect a train at a crossing and updates your navigation app to suggest a faster route.

The image highlights that computation takes place close to the devices means processing data directly on devices like cars, smartphones, or laptops instead of sending it to a central server. This makes the system faster and more efficient.

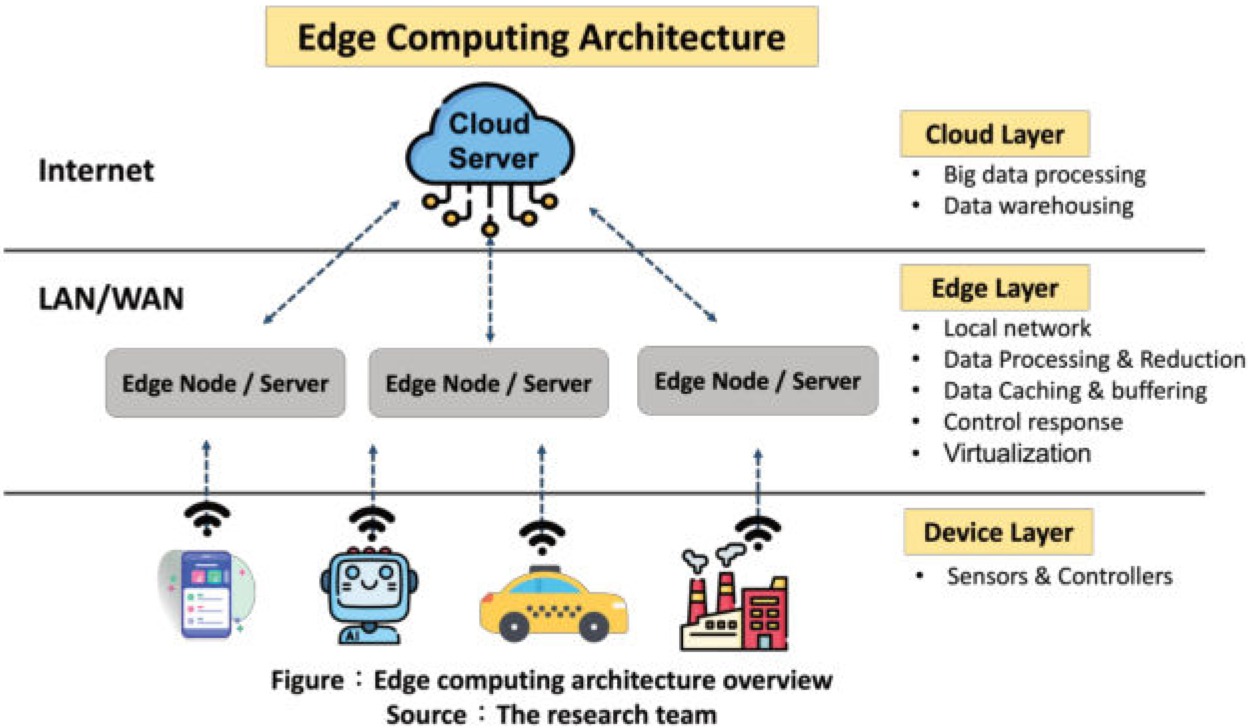


Fig 1.2.4 Edge Computing

1.2.4 5G Technology:

A **Cellular network** is a system of towers and antennas that provide wireless communication services, like mobile phone calls and internet access. 5G is the 5th generation of cellular network technology developed to make telecommunication networks faster.

With 5G, users can experience high-speed internet, which means faster mobile phone service, quicker downloads, and better connectivity between many devices. This improved speed and connectivity are helping to advance societies and enhance everyday experiences.

Example: With **5G**, you can stream a **1080p** YouTube video without buffering, offering smooth playback and high-quality resolution, unlike slower networks like 4G.

1.2.5 Artificial Intelligence:

AI can solve complex, real-world problems. Here's an explanation of each key problem AI addresses:

Decision-making: AI helps systems make decisions based on data. For example, self-driving cars use AI to make decisions about when to stop, go, or turn, just like a human driver would.

Learning: AI can learn from past experiences or data, similar to how humans learn. For instance, AI can improve its performance in a game by analyzing previous moves and adapting to new strategies.

Problem-solving: AI is used to find solutions to problems that are too complex or time-consuming for humans. For example, AI is used in medicine to analyze large amounts of patient data to diagnose diseases more accurately.

These capabilities allow AI to tackle challenges in fields like healthcare, finance, transportation, and more.

1.2.6 Robotics:

Robotics is a field of study within engineering and computer science that focuses on the design, construction, operation, and use of robots.

Robots are machines programmed to carry out specific actions, and they can work automatically or be controlled remotely. For example, in automobile manufacturing, robots are used to assemble car parts with high precision, making the process faster and safer.

Robots are used in areas like packaging, laboratory research, and home cleaning.

1.2.7 Distributed Applications:

Distributed Applications are programs that run on multiple computers at the same time to complete a task. This helps to distribute the work across several machines for processing, without overloading the single computer system.

For example, in **online banking**, several servers work together to process transactions. Similarly, **social networks** use distributed applications to manage data and posts, while **cellular networks** and **supermarket check-out systems** also rely on multiple computers to function smoothly.

Activity:

1. Can you think of an example where you've used an app that might involve distributed applications?
2. How would a distributed application help if a website had millions of users at once?

1.2.8 Blockchain Technology:

Blockchain is a system where information is stored in blocks that are linked together in a chain. Once information is added to a block, it cannot be changed or deleted. These blocks are connected to each other, forming a chain, which is known as a blockchain.

Blockchain is a type of database that ensures secure storage of digital information. It is commonly used in banks and financial services to record and share data securely, such as transactions. This technology helps to keep the information safe from tampering or unauthorized access.

Quick Tip:

Quick Tip: Blockchain ensures data security and transparency, making it ideal for use in industries like banking, where trust and accuracy are essential.

1.2.9 3D and Holographic Imaging:

3D and Laser Beam Technology uses laser beams to create three-dimensional images called holograms. These holograms look like real objects or animations, but they are digital projections created using laser light.

In **3D Holographic Imaging**, the hologram floats in space. Viewers can walk around it and see it from all angles, just like a real object. This technology is used in areas like entertainment, education, and advertising to create cool and interactive experiences.

Important: 3D means images that have depth, width, and height, while laser beams are focused light that help make clear and bright holograms. Together, they make virtual objects look real in space.

1.2.10 Virtual Reality:

Virtual Reality is a technology that makes computer-generated worlds that look and feel real. It creates a completely fake environment that users can interact with.
To experience virtual reality, users usually wear special helmets or goggles. Examples include video games and flight simulators, where users can explore and interact with a virtual world.

How it helps: Virtual reality helps in training, education, and entertainment by allowing users to practice skills in a safe, controlled environment or explore things they can't do in real life.
Fun Fact: Virtual reality lets you step into a whole new world, even though it's all made by a computer!

1.2.11 Computer-Assisted Translation (CAT):

Computer-Assisted Translation (CAT) is the use of computer software to help translate text from one language to another. This software ensures consistency across translations and helps reduce costs.
Computer-Assisted Translation (CAT) helps in education, literature, and personal tasks, improving translation speed and consistency. It's useful for students, writers, and organizations working with different communities.

Translation Memory (TM): CAT tools store previously translated text so they can reuse it later, helping keep translations consistent.

1.2.12 Data Analytics:

Data Analytics is the process of looking at large amounts of data to find patterns and trends. This helps people understand the information better and make decisions based on it.

It is used in many fields like education, healthcare, finance, and manufacturing. By analyzing data, organizations can improve their processes, predict future trends, and make smarter choices.

Activity:

Students can survey their classmates about their favorite food. If many students choose biryani, they can decide to serve biryani at the annual day event. This shows how data helps in making decisions!

1.2.13 Machine Learning:

Machine learning is a part of artificial intelligence (AI) that helps computers learn from data. Instead of being told what to do, the computer looks at past data and makes decisions or predictions based on that information.

Examples of machine learning include things like recognizing speech or images, translating languages, self-driving cars, Google Maps, and filtering out harmful software (malware). These applications use data to make tasks smarter and more efficient.

Table 1: The basic difference between "***Artificial Intelligence***" and "***Machine Learning***"

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)
Meaning	Making machines smart to do tasks like humans.	Teaching machines to learn from data.
Focus	Solving problems and making decisions.	Learning patterns from past data.
Examples	Siri, chatbots, recommendation systems.	Self-driving cars, spam filters, image recognition.

1.3 Computer Network:

1.3.1 Basic Terms of Network & Networking

Network:

A computer network is a system where ICT (Information and Communication Technology) devices are connected to share resources such as hardware (printers, storage) and software (applications). It enables communication and efficient data sharing.

Networking:

Computer networking is the process of connecting two or more computing devices to share data, resources, and services. It enables communication between devices like computers, servers, smartphones, and other hardware over wired or wireless connections.

1.3.2 Data Communication:

This is the process of exchanging information between devices in a network. For communication to happen, certain components are involved:

Sender: The person or device starting the communication.
Message: The information that is sent.
Medium: The path through which the message travels (e.g., cables, Wi-Fi).
Receiver: The person or device that receives the message.
Protocol: Protocols are a set of rules that ensure the sender and receiver communicate properly. Without protocols, devices wouldn't understand each other's messages.

1.3.3 Client-Server Architecture:

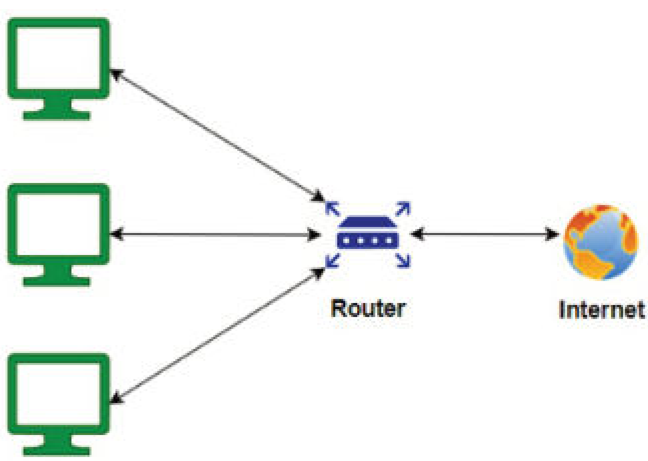
In a network, this architecture describes how devices communicate and share services:

- **Client:**
  A device or computer that requests services or data from a server. For example, when a student uses a computer in a school to access the school's online portal, the computer (the client) sends a request to the school's server.
- **Server:**
  A device or computer that provides the requested services or data to the client. In this case, the school's server processes the student's request and provides access to services such as class schedules, grades, or assignments.

1.3.4 Types of Computer Networks:

1.3.4.1 Local Area Network (LAN):

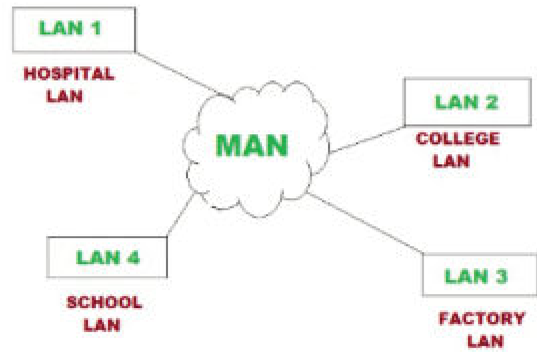
A **Local Area Network (LAN)** is a network of ICT devices connected within a small geographical area, such as a school, residence, or laboratory. It allows devices within that area to share resources like files, printers, and internet access. LANs typically provide fast and secure communication due to their limited size and scope.



Local Area Network - LAN

1.3.4.2 Metropolitan Area Network (MAN):

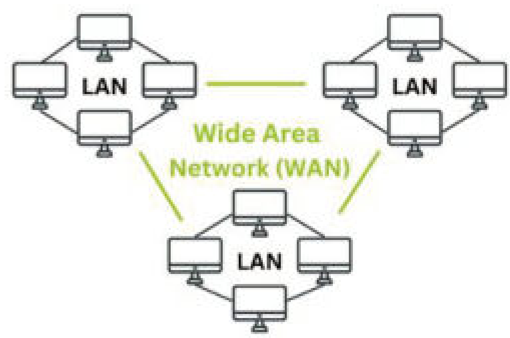
A **Metropolitan Area Network (MAN)** is a network that spans a larger geographical area than a LAN, but smaller than a WAN. It typically covers a city or a university campus with multiple buildings. MANs allow for the connection and communication between various LANs in a specific region.



Metropolitan Area Network - MAN

1.3.4.3 Wide Area Network (WAN):

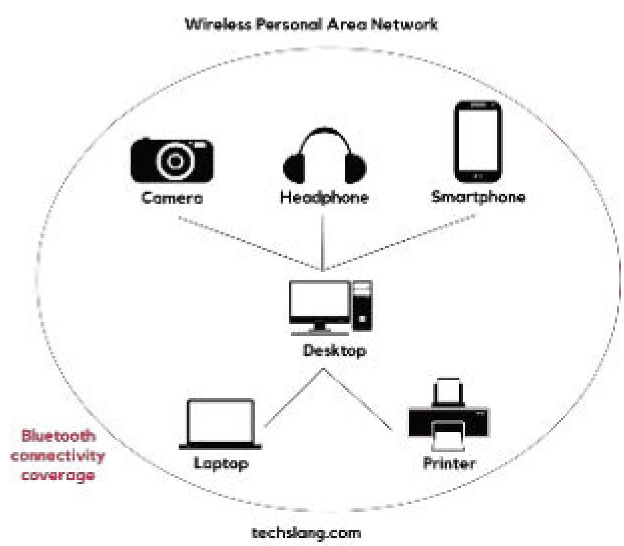
A **Wide Area Network (WAN)** is a large network of ICT devices that connects different smaller networks, including Local Area Networks (LANs), over a wide geographical area. WANs allow for communication between devices in different cities, countries, or even continents. An example of a WAN is a banking system that connects branches in multiple locations to share data.



Wide Area Network - WAN

1.3.4.4 Personal Area Network (PAN):

A **Personal Area Network (PAN)** is a network of electronic devices that connects within an individual's workspace. These devices typically communicate wirelessly and over short distances, typically less than 10 meters. PANs are commonly used to connect devices like smartphones, laptops, and Bluetooth-enabled devices.



Personal Area Network - PAN

1.3.4.5 Virtual Private Network (VPN):

A **Virtual Private Network (VPN)** allows users to securely connect to the internet by creating a private network over a public network. It encrypts the user's data and helps maintain privacy when accessing the internet. An example of a VPN is the secure connection between an ATM and the bank's central server.

Virtual Private Network - VPN

1.3.5 Software Networking Components and Their Uses:

1.3.5.1 Network Protocols:

In a data communication system, protocols play an essential role as they define the set of rules that both the sender and receiver follow to ensure effective communication and understanding of the transmitted data.

Example: TCP/IP (Transmission Control Protocol/Internet Protocol) is a common protocol that governs how data packets are transmitted over the internet, ensuring they reach the correct destination reliably.

1.3.5.2 Network Operating System (NOS):

In the client-server architecture, the server is a dedicated system that hosts and manages resources such as data, applications, and services. This often includes a Network Operating System (NOS) to manage network resources and services efficiently.

Example: Windows Server is a popular NOS that provides services like file sharing, printer management, and user authentication in business environments.

1.3.5.3 Network Management Software:

Is critical for monitoring, managing, and troubleshooting a network, ensuring it operates smoothly and securely.

Example: Solar Winds Network Performance Monitor is widely used software that helps IT teams monitor network performance, identify issues, and optimize resources effectively.

1.3.6 Hardware Networking Components and Their Uses:

1.3.6.1 NIC (Network Interface Card):

A NIC is a hardware component installed in devices like computers and printers, allowing them to connect to a network. It handles the sending and receiving of data within the network.

1.3.6.2 Modem:

A combined device for **modulation** and **demodulation** is called modem. A modem converts digital signals from a computer into analog signals for transmission over telephone or cable lines, enabling internet access. It also demodulates incoming analog signals back into digital form for the computer to process.

1.3.6.3 Hub:

A hub is a basic device used to connect multiple computers in a network. It sends data to all connected devices, which can reduce network efficiency as every device receives all the data, even if it's not intended for them.

1.3.6.4 Switch:

A switch is more advanced than a hub and connects multiple devices within a local area network (LAN). Unlike a hub, it sends data only to the specific device it is meant for, improving network performance and efficiency.

1.3.6.5 Router:

A router connects different networks, such as your home network to the internet. It directs data packets to their correct destination, ensuring that information reaches the right place.

1.3.6.6 Gateway:

A gateway serves as a bridge between two networks with different protocols, such as connecting a home network to the internet. It translates data formats so that communication between the networks is possible.

1.3.6.7 Access Point:

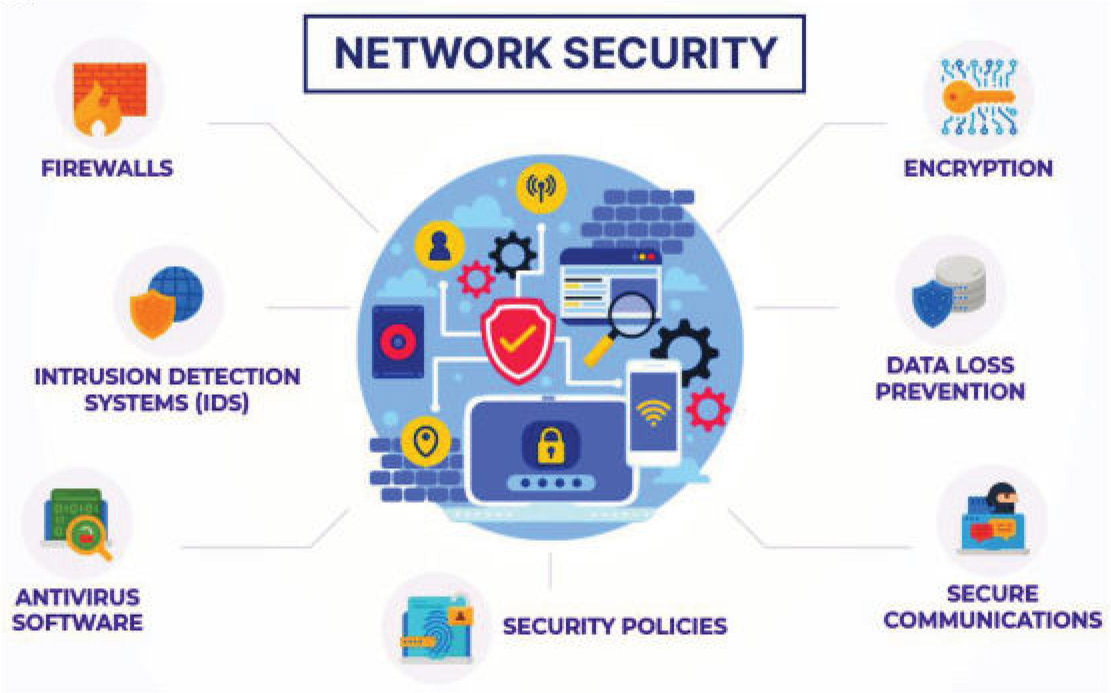
An access point creates a wireless network by allowing Wi-Fi-enabled devices, like laptops and smartphones, to connect to a wired network. It provides wireless access to the internet or other network resources.

1.3.7 Network Security Issues:

Network security is the practice of protecting a computer network from threats and unauthorized access by using tools like firewalls, encryption, and antivirus software to keep data and devices safe from cyberattacks.

1.3.7.1 Unauthorized Access:

When someone gains access to your network without permission, they can steal sensitive information, install malicious software, or disrupt services. Strong passwords, two-factor authentication, and regular software updates can help prevent unauthorized access.



1.3.7.2 Broken Components:

Broken components can lead to data loss or corruption. In a network, they can cause disruptions in communication between devices, making it difficult to access necessary resources. Regularly checking and maintaining network infrastructure can help avoid broken resources.

1.3.7.3 Viruses and Malware:

A virus or malware infection on one device can quickly spread to other devices on the network, causing widespread damage. Using robust antivirus software, practicing safe browsing habits, and being cautious with email attachments and downloads can help protect against these threats.

1.3.8 Firewalls and its Application:

Windows Firewall is a utility software that helps protect your computer by blocking harmful traffic and allowing safe connections. It acts as a security system that monitors and controls the flow of data in and out of a network, based on predefined security rules.

1.3.8.1 Applications of Firewalls:

- **Home Networks:** Protect personal devices from external threats.
- **Corporate Networks:** Secure sensitive company data and control employee internet usage.
- **Cloud Services:** Secure cloud-based applications and data from unauthorized access.

Example:

Windows operating systems have built-in security features to help protect users from online threats. One of the key security components is **Windows Firewall**, which is an essential tool for safeguarding your computer from malicious attacks.

1.3.9 Emerging Technologies in Network:

Emerging technologies in networking refer to new and advanced innovations that enhance the design, management, and security of networks. Technologies like 5G, IoT, and edge computing aim to make networks faster, more flexible, and efficient, meeting the increasing demand for data and connectivity.

1.3.9.1 5G Networks:

5G technology enhances network connectivity by providing faster data speeds, lower latency, and the ability to connect a vast number of devices simultaneously. This improvement supports more efficient communication and data transfer within a network.

- **High-Speed Internet:** 5G networks can handle more data at higher speeds, allowing for smoother streaming, quicker downloads, and better overall performance for all connected devices.
- **IoT Integration:** With 5G, the increased bandwidth and connectivity support a larger number of IoT devices, making it easier to manage and control them within a network.

1.3.9.2 Internet of Things (IoT):

The IoT creates a network of interconnected devices that communicate and share data. This allows for:

- **Automation and Efficiency:** IoT devices can perform tasks automatically and more efficiently, such as smart thermostats adjusting the temperature based on your preferences or smart fridges monitoring food supplies.

- **Enhanced Connectivity:** IoT devices use the network to exchange data and receive commands, enhancing their functionality and usability. For instance, a smart home network can connect lights, security cameras, and appliances, allowing them to work together seamlessly.

1.3.9.3 Edge Computing:

Edge computing processes data closer to where it is generated, reducing the need for data to travel long distances to centralized data centers. This approach benefits networks by:

- **Reduced Latency:** By processing data locally, edge computing minimizes the delay in data transmission, leading to quicker responses and real-time data analysis. This is crucial for applications like autonomous vehicles and smart traffic systems.

- **Efficient Bandwidth Usage:** Edge computing decreases the amount of data sent over the network, saving bandwidth and reducing congestion. This makes the network more efficient, particularly in environments with a high volume of data, like smart cities or industrial IoT setups.

Summary

- **Network** is connection of multiple devices to share data and resources.
- **Routers**: Devices that direct internet traffic between different networks.
- **Switches**: Devices that connect multiple devices within the same local network.
- **Modems**: Devices that convert digital data into analog signals for transmission and vice versa.
- **Network Interface Cards (NICs)**: Hardware components that allow devices to connect to a network.
- **Firewalls**: Security devices that monitor and control incoming and outgoing network traffic.
- **Fifth Generation (5G)**: A new wireless technology providing faster speeds and enhanced connectivity for mobile devices.
- **Internet of Things (IoT)**: A network of interconnected devices that can exchange data with each other.

- **Edge Computing**: A distributed computing framework that processes data closer to the source, reducing latency.
- **Artificial Intelligence (AI) and Machine Learning (ML)**: Technologies that allow machines to analyze data, make decisions, and improve over time.
- **Blockchain**: A secure, decentralized system used for recording transactions and managing data.

Exercise

1. Encircle the correct answers.

i. 5G technology improves network connectivity by providing:
a) Slower data speeds
b) Lower bandwidth
c) Faster data speeds and lower latency
d) Reduced number of devices connected

ii. IoT (Internet of Things) enables devices to:
a) work separately
b) connect and share data
c) only work on their own
d) do not need internet

iii. Main purpose of edge computing is:
a) To process data at a far distance
b) To process data closer to where it is generated
c) To reduce network speeds
d) To store data in the cloud

iv. In network, router:
a) connects devices within the local network
b) connects different networks and directs data to the correct destination
c) increases internet speed
d) connects wireless devices to the network

v. The main role of a firewall in network security is:
a) To block internet access
b) To monitor and control incoming and outgoing network traffic
c) To prevent data from being shared
d) To improve data transfer speeds

vi. **The main feature of IoT devices in a network is that:**
a) They are isolated from other devices
b) They exchange data and commands with other devices
c) They require manual operation
d) They cannot connect to the internet

vii. **The following is a result of broken resources in a network?**
a) Increased speed
b) Data loss or corruption
c) Improved communication between devices
d) Secure access to resources

viii. **In the network, edge computing helps reduce:**
a) Latency
b) Device connectivity
c) Bandwidth usage
d) Internet access

ix. **5G helps IoT devices by providing higher speeds and better connectivity for more:**
a) devices connected
b) Networks
c) speed
d) data transfer

x. **In a network, devices communicate more efficiently through:**
a) Hub
b) Router
c) Switch
d) Modem

2. Fill in the blanks with appropriate words.

i. _______ technology enables faster data speeds and lower latency, improving connectivity for a variety of devices.
ii. _______ is the practice of processing data closer to its source, which helps reduce latency and bandwidth usage in network communications.
iii. A _______ is a device that connects different networks and ensures that data is routed to the correct destination within a network.

iv. In a network, a _______ directs data traffic between multiple devices, ensuring efficient communication within a local area network (LAN).

v. _______ are devices that collect data from their environment and send it to IoT systems, enabling automation in smart homes and other industries.

vi. _______ is the technology that connects physical objects to the internet, allowing them to send and receive data for improved functionality.

vii. The main function of a _______ in a network is to monitor and control incoming and outgoing traffic to protect against unauthorized access.

viii. _______ systems are embedded in devices like washing machines, automating tasks and ensuring they function without manual intervention.

ix. _______ technology is expected to improve connectivity for more devices, enhancing applications like autonomous vehicles and smart cities.

x. The _______ is responsible for managing communication between devices in a wide area network (WAN), which connects LANs over large distances.

3. Provide the descriptive answers of the following questions.

i. What are the main components of a computer network, and how do they enable efficient communication?

ii. What security challenges do networks face, and what measures can be taken to protect sensitive information?

iii. How does IoT improve daily life, and can you provide an example of its application in healthcare?

iv. What is the role of sensors in IoT devices, and how do they enable remote control?

v. What is an embedded system, and how does it automate devices like washing machines?

vi. Identify three devices in your surroundings that use embedded systems and explain their tasks.

vii. What is edge computing, and how does it reduce delay in data processing?

viii. Describe how edge computing can be applied to traffic management systems.

ix. How does 5G technology enhance connectivity, and what industries benefit the most from it?

x. What are the potential risks associated with 5G implementation, and how can they be mitigated?

Important points to remember

- **Emerging Technologies** are new technologies that are still being developed or are just starting to be used. These can affect many areas, like artificial intelligence, blockchain, and quantum computing.
- **Emerging Technologies in Networking** focus on new technologies that improve how networks work, helping with communication, security, and the way networks are built and run.
- **Networking Components**: The different parts of a network, like routers, switches, and cables, that allow devices to communicate.
- **IP Address:** A unique number that helps to identify a computer or device on the internet.
- **Latency:** The delay or wait time it takes for data to travel across the internet.
- **VPN (Virtual Private Network):** A tool that keeps your internet activity private and secure by creating a safe connection.
- **Bandwidth**: The amount of data that can be transmitted over a network in a given time.
- **Firewall**: A security system that monitors and controls incoming and outgoing network traffic, protecting the network from unauthorized access.

Instruction for Teachers:

- **IoT Video or Presentation**
  Show videos or slides illustrating real-world IoT examples like smart homes or self-driving cars to highlight the technology's impact.
- **Embedded Systems in Common Devices**
  Use everyday devices (e.g., switch/router, smartphone) to demonstrate how embedded systems are integrated and function.
- **AI in Social Apps**
  Explain how AI improves social media apps with personalized recommendations, facial recognition, and chatbots.
- **Blockchain Simulation with Paper**
  Create a classroom activity using colored paper blocks to demonstrate blockchain concepts like transparency and security.
- **Simple LAN Activity**
  connect two devices (laptop/smartphone) to the same Wi-Fi network and show how they communicate and share resources within a local network.